

Introduction

This publication provides a comprehensive report on the Economics examination in the 2016 Hong Kong Diploma of Secondary Education Examination. It covers various aspects of the examination, including the objectives, contents and format of the examination, the question papers, the marking schemes, and the chief examiners' comments on candidates' performance in the live examination. Some statistics are also included.

Through this publication, we hope to provide useful information and feedback to assist teachers in their teaching of the subject and students in their learning. Similar publications are also available for other subjects in the 2016 HKDSE. Besides, a series of publications and resource materials related to the HKDSE are also available for reference to schools, candidates and the public. These include information related to examination administration, assessment requirements, examination-related statistics and feedback provided to schools and other stakeholders. For details, please visit the Authority's website (http://www.hkeaa.edu.hk/en/HKDSE/info_corner/hkdse_publications_materials/).

We would like to take this opportunity to express our gratitude to all the examination personnel who were involved in the development of question papers and the assessment of candidates' performance. Without their contributions, it would not have been possible for us to conduct the examination smoothly.

The Examination

Assessment Objectives

The objectives of this assessment are to evaluate students' abilities and achievements in the following:

1. to know and understand fundamental economic concepts and theories;
2. to apply such concepts and theories to explain real world situations, especially the Hong Kong economy;
3. to understand and interpret economic information presented in various forms;
4. to acquire the basic tools of economic analysis;
5. to analyze economic issues;
6. to evaluate arguments, proposals and policies from different perspectives and make informed judgments; and
7. to present ideas clearly, and in a well-reasoned manner, illustrating answers with diagrams and examples.

Synopsis of curriculum contents

For details, please refer to the Economics Curriculum and Assessment Guide (Secondary 4 to 6) jointly prepared by the Curriculum Development Council and the Hong Kong Examinations and Assessment Authority.

Compulsory Part

1. Basic Economic Concepts
2. Firms and Production
3. Market and Price
4. Competition and Market Structure
5. Efficiency, Equity and the Role of Government
6. Measurement of Economic Performance
7. National Income Determination and Price Level
8. Money and Banking
9. Macroeconomic Problems and Policies
10. International Trade and Finance

Elective Part

Elective Part 1: Monopoly Pricing, Anti-competitive Behaviours and Competition Policy; OR
Elective Part 2: Extension of Trade Theory, Economic Growth and Development

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2016

ECONOMICS PAPER 1

8.30 am – 9.30 am (1 hour)

INSTRUCTIONS

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
2. When told to open this book, you should check that all the questions are there. Look for the words '**END OF PAPER**' after the last question.
3. All questions carry equal marks.
4. **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
6. No marks will be deducted for wrong answers.

Not to be taken away before the
end of the examination session

There are 45 questions in this paper. Choose the **BEST** answer for each question.

1. Which of the following statements about economic goods is correct?
 - A. Economic goods may not be scarce.
 - B. People compete for economic goods.
 - C. People prefer economic goods to free goods.
 - D. People compete for both economic goods and free goods.
2. During the Hong Kong Computers and Communications Festival, the organisers distribute USB flash drives to the participants as gifts. To the organisers, these USB flash drives are _____.
 - (1) economic goods
 - (2) producer goods
 - (3) consumer goods
 - A. (2) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (1) and (3) only
3. Mr Ng, a Hong Kong citizen, is going to travel to Tokyo for leisure. Which of the following will decrease his cost of having the trip?
 - A. A typhoon is approaching Tokyo right before his trip.
 - B. His boss is going to treat all the staff to a free buffet during his leave.
 - C. The scenery in Tokyo is better than he expected.
 - D. Japanese Yen depreciates against other currencies.
4. Which of the following statements about interest are correct?
 - (1) Interest is the compensation for deferring consumption.
 - (2) Interest exists because people are impatient to wait.
 - (3) Interest exists in a barter economy.
 - A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

5. Marlene had bought a cookery book from a bookstore. Her friend, Kitman, said to her, "You have private property rights to this cookery book." Which of the following rights would Marlene have after the purchase of the book?

- (1) To follow the recipes in the book in preparing a dinner for Kitman
- (2) To scan and upload the whole book to her blog
- (3) To leave the book on the bookshelf without reading it
- (4) To lend the book to Kitman and ask for a dinner in return

- A. (1), (2) and (3) only
- B. (1), (2) and (4) only
- C. (1), (3) and (4) only
- D. (2), (3) and (4) only

Study the following information and answer Questions 6 and 7:

A school is a production unit which provides educational services. The following are people and objects that are usually found at a school:

- (1) the school building
- (2) teachers
- (3) students
- (4) private cars owned by teachers
- (5) sanitisers (洗手液) in the toilets

6. Which of the above items are capital of the school?

- A. (1) and (4) only
- B. (1) and (5) only
- C. (2) and (3) only
- D. (1), (4) and (5) only

7. In the long run, which of the above items are variable factors of production of the school?

- A. (1) and (5) only
- B. (1), (2) and (5) only
- C. (2), (3) and (4) only
- D. (1), (2), (3), (4) and (5)

8. The labour supply of a country will increase if

- A. more training is provided for the existing labour force.
- B. the legal minimum working age is lowered from 16 to 15.
- C. foreign direct investment increases.
- D. the government increases the salaries tax rate.

9. Which of the following is the most mobile in terms of occupational mobility?

- A. a professional basketball player
- B. a licensed plumber (水喉匠)
- C. a graduate trainee in a law firm
- D. a clerk in an accounting firm

10. Which of the following is most likely to be paid on a piece-rate basis?

- A. a newspaper columnist
- B. a primary school teacher
- C. a fireman
- D. a bus driver

11. Hong Kong Television Network Limited (HKTV) has launched an online shopping platform "HKTV Mall". This expansion

- (1) helps spread business risk.
- (2) ensures a higher profit for the company.
- (3) extends its brand name to other markets.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

12. The following table shows the cost-output relationship of a price-taker.

Output (units)	Average cost (\$)
2	20
3	21
4	22
5	23
6	24
7	25

Suppose the market price is \$25. What are the profit-maximising output and profit respectively?

- A. 4 units and \$0
- B. 4 units and \$12
- C. 7 units and \$0
- D. 7 units and \$12

13. Which of the following situations would lead to an increase in the demand for private housing?

- A. The price of private housing decreases.
- B. The supply of public housing increases.
- C. People expect the interest rate for mortgage loans to increase later.
- D. People expect the price of private housing to increase later.

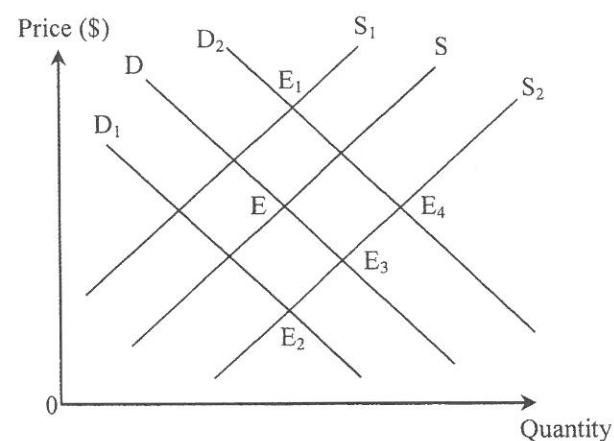
14. The demand for _____ can be regarded as a derived demand for _____.

- A. coffee beans coffee
- B. selfie sticks (自拍桿) digital cameras
- C. chicken wings chicken legs
- D. newspapers printing paper

15. A decrease in the cost of producing Good X results in a 5% and 10% change in its price and quantity transacted respectively. Which of the following best explains the above changes?

- A. Good X is a durable good.
- B. There are no close substitutes for Good X.
- C. Firms producing Good X have excess capacity in production.
- D. The factors of production for Good X are not easily available.

16. The diagram below shows the supply and demand curves of taxi services in Hong Kong. The original equilibrium is at E.



New apps for smartphone such as "Uber" have been developed that enable some private car owners to provide the general public with personalised point-to-point transport service. At the same time, the fuel price has decreased significantly. The new equilibrium point will be at _____.

- A. E₁
- B. E₂
- C. E₃
- D. E₄

17. The demand for construction workers in Hong Kong will increase if

- A. the legal procedures for urban re-development are simplified.
- B. wage rates of construction workers decrease.
- C. the government provides more allowances for training programmes for the construction industry.
- D. there are fewer job opportunities in the construction industry in Macau.

18. Which of the following is **INCONSISTENT** with the law of demand?

- A. When the government imposes an ad valorem tax (i.e. the tax is a certain percent of the price) on red wine, the ratio of high-quality red wine sold to low-quality red wine sold remains unchanged.
- B. A larger quantity of expensive fruits is sold in a faraway market than a market near the farm.
- C. Price of residential flat and quantity transacted are positively related during an economic boom.
- D. An increase in the monthly rental of parking spaces will lead to a decrease in the average quality of vehicles that use the parking spaces.

19. Refer to the following demand and supply schedules of a good.

Price (\$)	40	50	60	70	80	90
Quantity demanded (units)	14	12	10	8	6	4
Quantity supplied (units)	6	8	10	12	14	16

Which of the following statements is correct when the government provides a per unit subsidy of \$20 on the good?

- A. The total revenue including the subsidy is \$840.
- B. The total subsidy granted by the government is \$200.
- C. Within the range of the original and new market prices, the elasticity of demand is smaller than one.
- D. The change in total revenue of its complements is uncertain.

20. Which of the following statements about monopoly are correct?

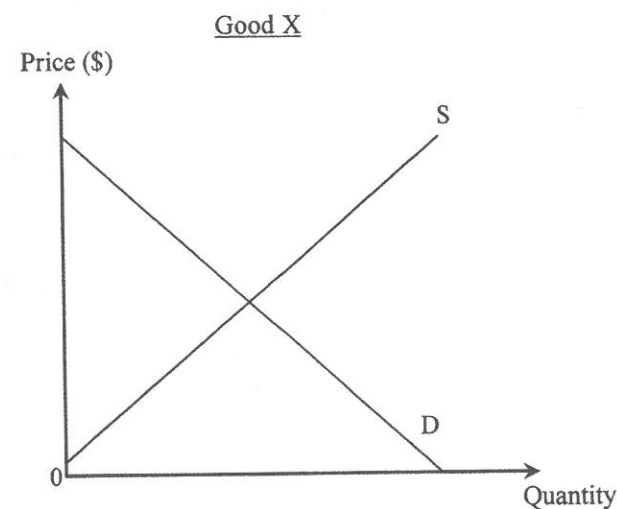
- (1) A monopolist sets the market price.
- (2) A natural monopoly arises when no firms are permitted to enter the market.
- (3) There are substitutes for the products provided by a monopolist.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

21. In which of the following situations is efficiency attained?

- A. Total social benefit is equal to total social cost.
- B. Producer surplus is equal to consumer surplus.
- C. The difference between marginal social benefit and marginal social cost is zero.
- D. The consumer surplus of all consumers in the market is maximised.

22. The following diagram shows the supply and demand curves of Good X.



Good X and Good Y are in competitive supply. A rise in the demand for Good Y will lead to _____ in the consumer surplus and _____ in the producer surplus in the market of Good X.
(Assume the supply and demand curves of Good X shift in a parallel manner.)

- A. an increase an increase
- B. an increase a decrease
- C. a decrease an increase
- D. a decrease a decrease

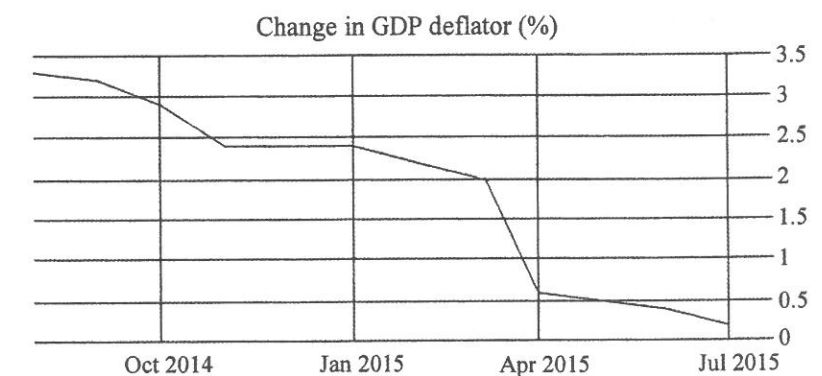
23. Consider the following Gross Domestic Product (GDP) data about a country.

GDP components	Million (\$)
Private consumption expenditure	380
Government consumption expenditure	450
Gross domestic fixed capital formation	230
Decrease in stock	50
Subsidies	30
Depreciation	40
Total exports	130
Total imports	170
Direct taxes	30

The GDP at factor cost is \$ _____ million.

- A. 960
- B. 970
- C. 1 000
- D. 1 100

24. Study the diagram below.



The diagram shows that during the period from October 2014 to July 2015,

- A. the nominal GDP was falling.
- B. the purchasing power of money was increasing.
- C. the growth rate of real GDP was higher than that of nominal GDP.
- D. the nominal interest rate was higher than the realised real interest rate.

25. Which of the following will reduce the unemployment rate of a country?

- A. Some university graduates join the labour force.
- B. Some factories move to a neighbouring country with lower labour cost.
- C. Some unemployed workers give up seeking jobs.
- D. The government increases unemployment benefits.

26. The aggregate demand of a country will drop if

- A. the price level of the country rises.
- B. the currency of its major trading partner appreciates.
- C. the government spending on social welfare rises.
- D. the required reserve ratio rises.

27. If the government builds new infrastructures, there will be an increase in

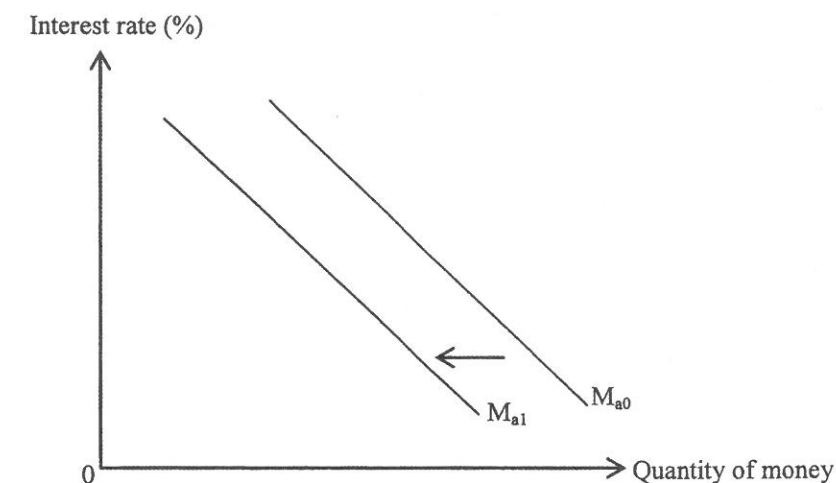
- (1) employed population during the construction period.
- (2) the price level during the construction period.
- (3) potential output after the construction is completed.
- (4) the price level after the construction is completed.

- A. (1) and (2) only
- B. (3) and (4) only
- C. (1), (2) and (3) only
- D. (1), (2), (3) and (4)

28. The 'multiple entry permit' (一簽多行) for the permanent Shenzhen residents to visit Hong Kong has been replaced by a 'one trip per week cap' (一週一行). This would reduce Hong Kong's _____ and _____.

- A. aggregate demand output level in the long run
- B. aggregate demand price level in the long run
- C. short-run aggregate supply output level in the short run
- D. long-run aggregate supply output level in the long run

29. Refer to the following diagram.



Which of the following would lead to the above change in the asset demand for money curve?

- A. The stock market becomes less volatile.
- B. The general price level is expected to fall continuously.
- C. The central bank raises the discount rate.
- D. There is a fall in disposable income of households.

30. Refer to the following diagram.



Ken read the above advertisement and bought a new mobile phone by installments. In this payment, which of the following functions did money perform?

- (1) store of value
- (2) unit of account
- (3) standard of deferred payment

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

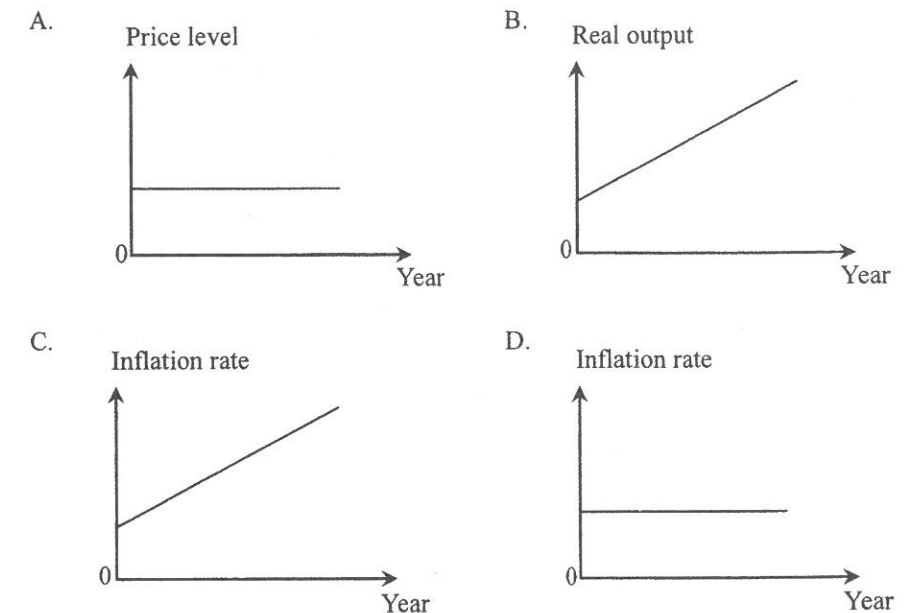
31. Refer to the following balance sheet of a banking system. Its excess reserve is \$125 million.

Assets (\$ million)		Liabilities (\$ million)	
Reserves	250	Deposits	500
Loans	250		

Which of the following statements is **INCORRECT**?

- A. The required reserve ratio is 25%.
 B. The maximum amount of loans is \$750 million.
 C. The banking system can expand its total deposit to \$1 000 million.
 D. The actual banking multiplier is 4.
32. Winnie withdraws HK\$500 000 from her savings account with a licensed bank. She puts half of it into her current account and deposits the other half at a deposit-taking company. Which of the following shows the immediate effects on the money supply of Hong Kong?
- A. M1 increases and M2 decreases.
 B. M1 increases and M3 decreases.
 C. Both M1 and M2 decrease.
 D. Both M2 and M3 decrease.
33. In a fractional reserve banking system, the actual deposit created is often below the maximum possible deposit created. The gap between these two amounts will be narrowed when
- A. the popularity of electronic payment increases.
 B. the central bank tightens the restrictions imposed on mortgage loans from banks.
 C. the central bank buys bonds from the public.
 D. the central bank reduces the legal reserve ratio for banks.

34. According to the classical quantity theory of money, which of the following diagrams is correct when money supply is growing at a constant percentage each year?

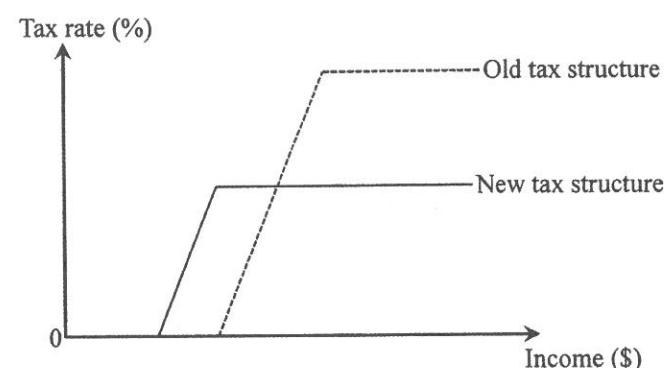


35. Mr Lam earns \$1 000 000 as a mobile app designer of an IT firm in Hong Kong. Also, he earns a \$50 000 dividend by holding shares issued by a local company. He buys a second-hand racing car and flies to the Maldives to spend his vacation through the Hong Kong International Airport.

Based on the above information, Mr Lam needs to pay _____ to the Hong Kong government.

- (1) salaries tax
 (2) dividend tax
 (3) first registration tax
 (4) air passenger departure tax
- A. (1) and (4) only
 B. (2) and (3) only
 C. (1), (2) and (4) only
 D. (1), (3) and (4) only

36. The following diagram shows the changes in the salaries tax structure of an economy.



Compared to the old tax structure, the new tax structure will _____.

- A. result in a more even income distribution
 B. reduce the work incentive of all workers
 C. broaden the tax base
 D. lead to an increase in total tax revenue
37. Which of the following people will gain in times of unexpected deflation?
- (1) holders of i-bonds issued by the Hong Kong Government
 (2) flat owners repaying floating rate mortgage loans
 (3) pensioners receiving fixed pensions
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)
38. If the government simultaneously increases its expenditure and income tax by the same amount, the effect on the aggregate output is _____.
- A. contractionary
 B. neutral
 C. expansionary
 D. indeterminate
39. Suppose Country A initially operates at the full-employment equilibrium and the fiscal budget of the government is balanced. In the short run, an economic recovery of Country A's trading partners will result in _____ and _____ of Country A.
- A. an inflationary (output) gap a fiscal surplus
 B. an inflationary (output) gap a fiscal deficit
 C. a deflationary (output) gap a fiscal surplus
 D. a deflationary (output) gap a fiscal deficit

40. Under which of the following situations may a fiscal deficit and a trade deficit be eliminated at the same time?

- (1) a decrease in nominal interest rate
 (2) a depreciation of the economy's currency
 (3) an increase in income tax

- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

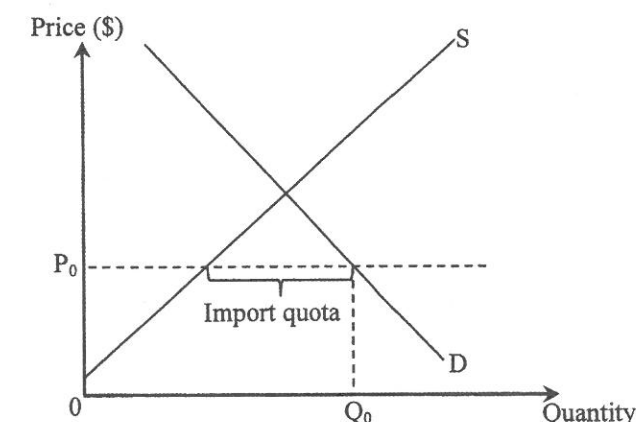
41. The following table shows the amount of toys or cars that Country A and Country B can produce with the same amount of resources.

	Toys		Cars
Country A	50	or	80
Country B	20	or	60

The transportation cost for 1 unit of toy is 0.4 units of car and it is borne by the importing country. Which of the following is the mutually beneficial terms of trade between Country A and Country B?

- A. 1 unit of toy = 1.6 units of car
 B. 1 unit of toy = 2 units of car
 C. 1 unit of toy = 2.6 units of car
 D. 1 unit of toy = 3 units of car

42. Refer to the following supply-demand diagram about Good X in a small country.



The country has imposed an import quota on Good X and the price and quantity transacted are P_0 and Q_0 respectively. If the demand for Good X in the country increases,

- (1) the import volume will increase.
 (2) the price of Good X in the country will increase.
 (3) the quantity sold of Good X in the country will increase.

- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

43. The following table shows the information about the balance of payments of a country:

	\$ million
Visible trade	20
Invisible trade	X
Capital inflow	70
Capital outflow	50

Suppose the foreign exchange reserves of the country increase by \$30 million. What is the value of X?

- A. -70
B. -10
C. +10
D. +70
44. Suppose the countries in a certain region are competitors of exporting goods to the U.S.. The currencies of the countries in the region except that of Country X depreciate against the US dollar. As a result, the volume of Country X's exports to the U.S. will
- A. decrease.
B. increase.
C. remain constant.
D. be indeterminate.

45.

Date	Exchange Rate
15 February 2015	US \$1 = € 0.8773
15 August 2015	US \$1 = € 0.8965

Under the linked exchange rate system of Hong Kong, the above change in the exchange rate of Euro (€) to the US dollar would lead to _____.

- A. an increase in the number of European tourists to Hong Kong
B. an increase in the total value (in terms of Hong Kong dollar) of European exports to Hong Kong
C. a decrease in the volume of Hong Kong exports to the U.S.
D. a decrease in the total value (in terms of Hong Kong dollar) of Hong Kong exports to Europe

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

ECONOMICS PAPER 2
Question-Answer Book

10.15 am – 12.30 pm (2 hours 15 minutes)
This paper must be answered in English

INSTRUCTIONS

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7, 9 and 11.
- (2) This paper consists of three sections, A, B and C. Answer ALL questions in sections A and B. Attempt any ONE question in section C.
- (3) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (4) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.
- (5) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

Please stick the barcode label here.

Candidate Number



Section A (50 marks)

1. Apple Inc. launches a new model of smartphone. In order to buy the new model, some people spend hours in trying to login the website of Apple Inc.. If the resale value of the new model falls, the opportunity cost of buying the new model through the website of Apple Inc. will decrease. Do you agree? Explain your answer. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. Sure-Star Education Group is a tutorial centre for secondary school students. The Group can issue shares and bonds to the general public to raise capital.
- (a) Name the type of business ownership of the Group. (1 mark)
- (b) From the viewpoint of owners, explain **ONE** advantage and **ONE** disadvantage of issuing shares over bonds. (4 marks)
- (c) The average cost of providing the services decreases when the Group operates more tutorial classes in different districts. Give **THREE** reasons to explain why its average cost decreases. (3 marks)

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Answers written in the margins will not be marked.

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3. The table below shows the input-output relationship of a firm.

Machine (units)	Labour (units)	Average output of labour (units)
2	1	10
2	2	12
2	3	14
2	4	14
2	5	13

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (6 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. These years many customers like to take photos of the food they have in restaurants and share them on social media. However, some high-end restaurants do not allow their customers to take photos. Explain how the “photo-taking-and-sharing” behaviour of the customers may involve externality. (3 marks)

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Answers written in the margins will not be marked.

5. Explain, with **ONE** reason, why the short-run aggregate supply curve is upward-sloping. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. Below are the employment data of a country.

Employed population	3 800 000
Underemployed population	800 000
Unemployed population	200 000

- (a) Calculate the unemployment rate of the country. (2 marks)
- (b) Suggest **THREE** losses to the society caused by unemployment. (3 marks)
- (c) Suppose the country is in the phase of recovery. Apart from the reduction in unemployment rate, give **TWO** general economic phenomena during recovery. (2 marks)

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Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. The input requirements in the production of one unit of clothing and one unit of toys are 6 units and 2 units respectively in Country A. Country A and Country B trade at an exchange ratio of 2 units of clothing to 11 units of toys.

- (a) Which good would Country A export? Explain. (3 marks)
- (b) Suppose a transportation cost of 2 units of toys is required to exchange for 1 unit of clothing and the cost is borne by Country A. Calculate the gain per unit of export for Country A. (3 marks)

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Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. The following is the balance sheet of a banking system.

Assets (\$million)		Liabilities (\$million)	
Reserves	1 000	Deposits	4 000
Loans	3 000		

Suppose the public always holds \$500 million of cash and the banking system does not hold excess reserves.

(a) Calculate the monetary base and money supply. (2 marks)

The central bank increases the required reserve ratio to 50%.

(b) Calculate the change in monetary base and the change in money supply. Show your working. (4 marks)

(c) (i) Why is money supply often a few times more than monetary base? (3 marks)

(ii) Under what situation will money supply be equal to monetary base? (1 mark)

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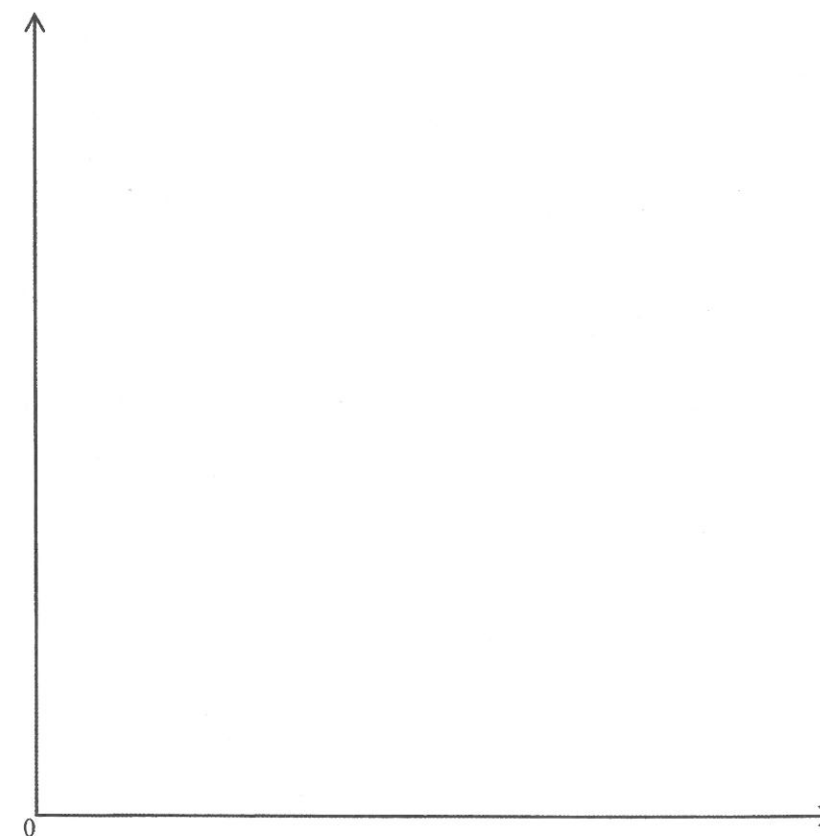
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9. Thousands of farmers of the European Union held street protests to urge the government to support the price of their dairy products which were affected by the following events.

- The European Union abolished the effective quota on the production of dairy products.
- Russia imposed an embargo on the import of dairy products from the European Union.

Use ONE supply-demand diagram to illustrate the effect of the above two events on the price of the dairy products of the European Union. (4 marks)



END OF SECTION A

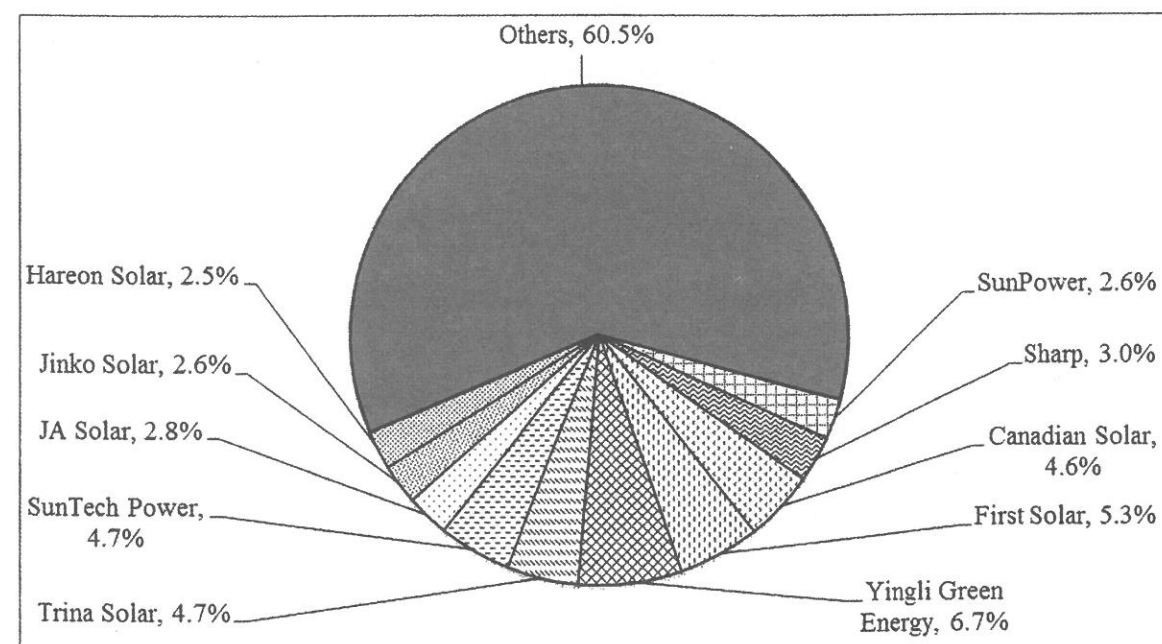
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Answers written in the margins will not be marked.

Section B (54 marks)

10. Solar energy becomes an increasingly important energy source in the sunny western provinces, like Gansu and Xinjiang, in China.

The diagram below shows the market shares of Top 10 solar photovoltaic (PV) module manufacturers in 2012.



- (a) What was the market structure of the above solar PV module market? List **TWO** features of this market structure. (3 marks)

Solar panels are installed atop the roofs of many residential buildings to generate solar energy in those western provinces.

- (b) Explain to which type of factors of production the sunshine belongs in the production of solar energy. (2 marks)

The European Union has imposed an anti-dumping tariff on the Chinese produced solar panels.

- (c) Under what condition will the producers bear a greater tariff burden than the buyers? Illustrate your answer with a supply-demand diagram. (7 marks)

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11. Medical service is one of the basic human needs.

Source A: a conversation outside a public clinic in Hong Kong

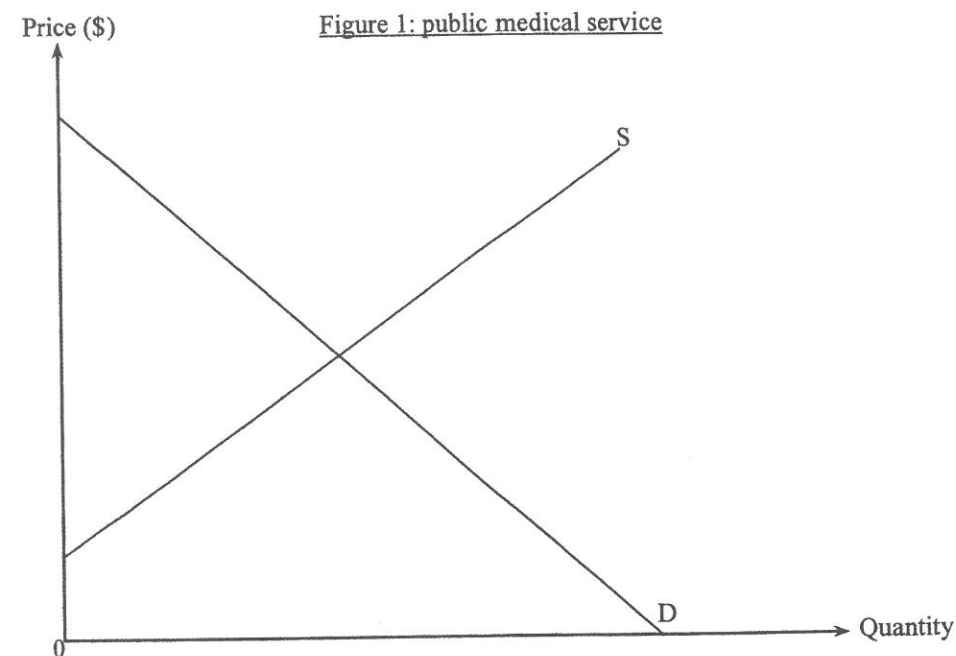
Son: Dad, I feel very ill.
 Father: Oh no! There is a long queue waiting.
 Son: Why don't we go to a private doctor?
 Father: The fee charged by public clinics is only \$45, much lower than that charged by private doctors.

- (a) Refer to Source A. When a public clinic provides medical service for one patient, the resulting rise in gross domestic product will be \$45. Do you agree? Explain. (3 marks)
- (b) With reference to Source A, explain with the aid of Figure 1 how the price set by public clinics may lead to economic inefficiency in the provision of public medical service. (6 marks)
 (Assume the supply curve in Figure 1 fully reflects the marginal social cost of public medical service.)

Source B: an extract from the Hospital Authority webpage

The Hospital Authority is committed to providing community-based primary care services. Our target patients include the elderly, low-income individuals, and patients with chronic diseases.

- (c) With reference to Source B, explain how the services provided by the Hospital Authority can enhance equity. (2 marks)
- (d) Many people argue that public clinics should raise the fee to reduce the waiting time. Evaluate this argument in terms of equity and economic efficiency. (4 marks)

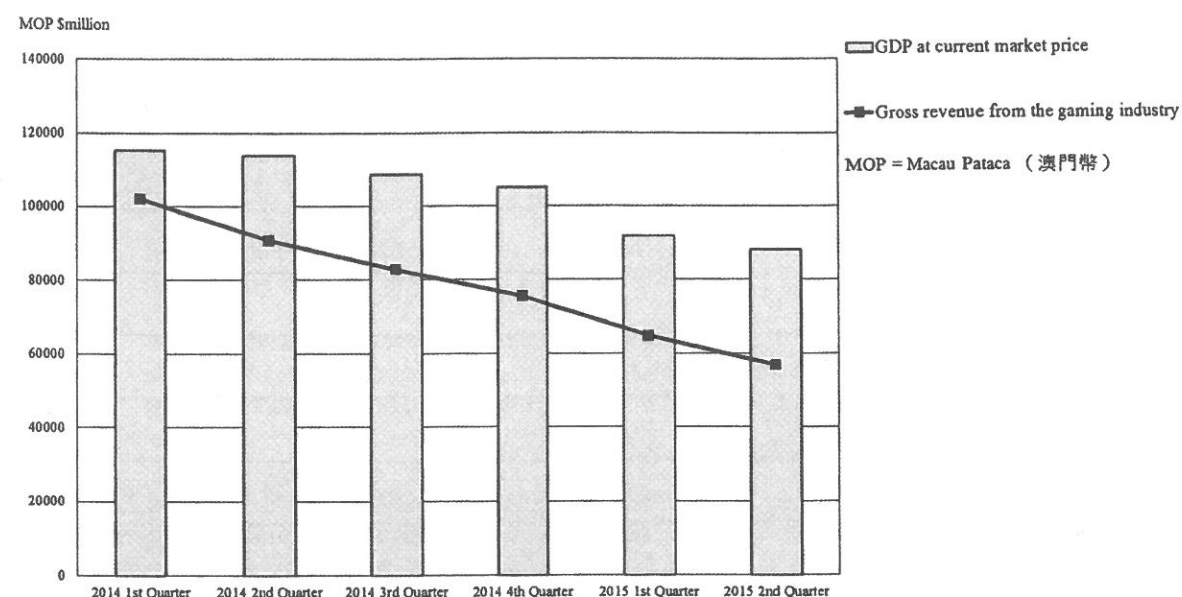


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12. Gaming industry (博彩業) is a major sector in Macau's export of services. The following diagram shows the revenue from the gaming industry and the aggregate output of Macau.



- (a) With reference to the above diagram, describe and explain the trend of the importance of the gaming industry relative to the aggregate output of Macau. (2 marks)

The Macau government implements a cash-sharing scheme, under which every resident receives cash from the government.

- (b) With the aid of an AS-AD diagram, explain how the cash-sharing scheme would affect output and price levels in the short run. (6 marks)

The recent decline in the gaming industry results in a drop in tax revenue. Hence, the government considers reducing its spending on investment but still keeping the cash-sharing scheme. However, an economist suggests that the government should suspend the cash-sharing scheme and allocate the sum of money to investment instead.

- (c) Discuss the suggestion of the economist by giving arguments **FOR** and **AGAINST** it. (5 marks)

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13. The airport corporation of Country H is a public corporation which hires different people to operate the airport. For example, there are security guards, technicians, engineers and financial managers.

(a) What economic principle is illustrated by this practice? Suggest **THREE** reasons why the application of this principle can increase labour productivity. (4 marks)

(b) The airport corporation plans to build an additional runway. There are two proposals to finance the runway.

Proposal A: the government increases the income tax
Proposal B: the airport corporation charges a lump-sum fee on every departing traveller

Discuss the effects of the two proposals on

(i) the current account in the balance of payments of Country H, and

(ii) the average flight distance for departing travellers of Country H.

(10 marks)

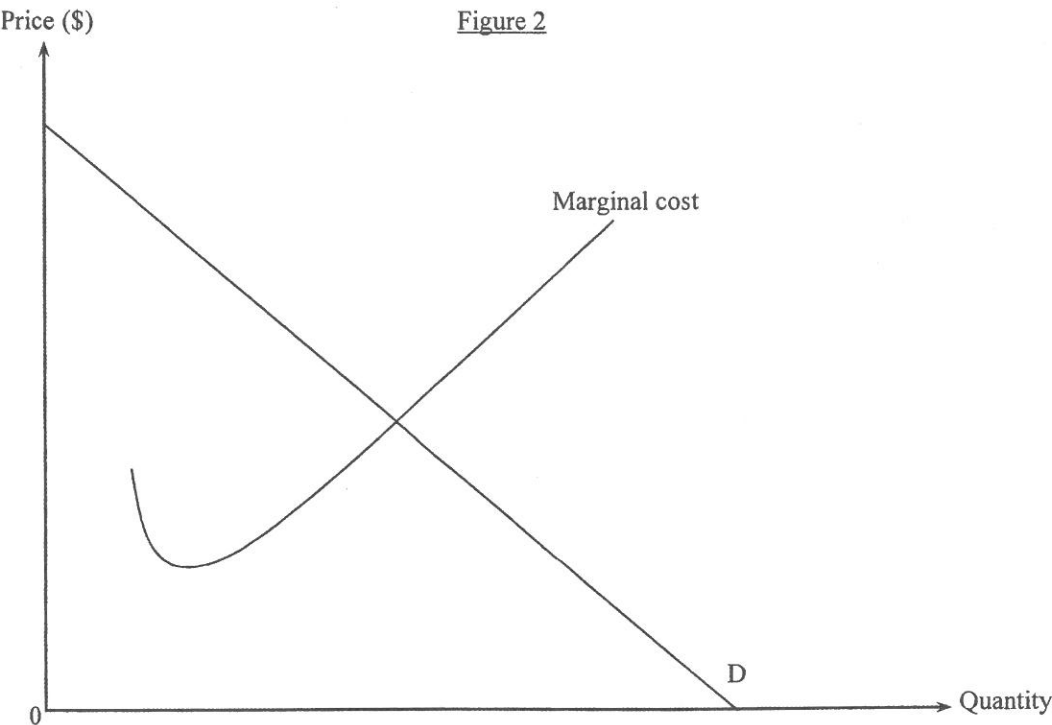
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END OF SECTION B

Section C (16 marks) Answer any ONE question.

14. Cheung Chau, an outlying island of Hong Kong, is accessible only by ferry. There is only one franchised ferry company providing public transport service between Central and Cheung Chau.

- (a) Suppose the ferry company adopts simple monopoly pricing.
- (i) Indicate the output and price as Q_m and P_m respectively in Figure 2. (2 marks)
- (ii) With the aid of Figure 2, explain why such a pricing arrangement would lead to a deadweight loss. (2 marks)



- (b) The fuel price has dropped in recent years.
- (i) How would the reduction of fuel price affect the company's marginal cost? (1 mark)
- (ii) Suppose the ferry company produces the **SAME** quantity and charges the **SAME** price after the fall in fuel price. Explain whether economic efficiency would be affected. Illustrate your answer in Figure 2. (3 marks)

(c) The table below shows the fare of ferry service from Central to Cheung Chau.

Adult fare	\$13.20
Child fare	\$6.60

Explain whether the above arrangement constitutes price discrimination. (3 marks)

- (d) Many restaurants in Cheung Chau serve seafood dishes. Some of them raise the prices by about 15% on public holidays simultaneously. A student commented, "The restaurants engage in anti-competitive behaviour. They violate a conduct rule under the Competition Ordinance!" His Economics teacher replied, "There may be an alternative explanation for the rise in prices."
- (i) Explain under what situation the student's comment may be correct. (3 marks)
- (ii) Give **ONE** reason to support the teacher's reply. (2 marks)

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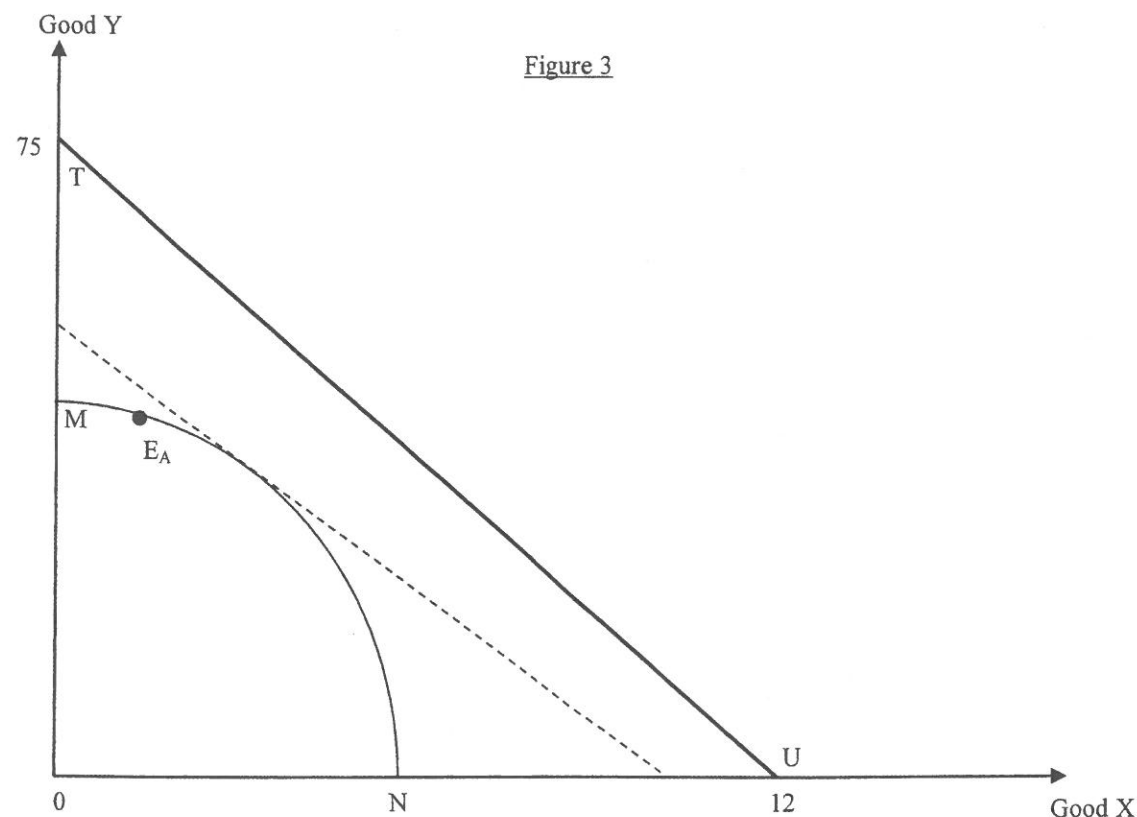
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15. In Figure 3, MN and TU are the production possibility frontiers of Country A and Country B respectively.



- (a) Explain why the production possibility frontier of Country A is concave to the origin. (1 mark)

The production point of Country A without trade is E_A , at which the domestic cost of production of 1 unit of Good X is 4 units of Good Y.

- (b) With reference to the domestic cost of production in both countries, explain in which good Country A has a comparative advantage. (3 marks)

Suppose the slope of the dotted line is the world price of Good X.

- (c) Indicate in Figure 3:

- the production point of Country A after trade as P_A ,
- the consumption point of Country A as C_A given that Country A consumes the **SAME** amount of Good X after trade, and
- the volume of import and the volume of export of Country A.

(4 marks)

- (d) Country B has a technological improvement which helps double its productivity on Good X and Good Y. Explain whether the answer in part (b) will change. (2 marks)

- (e) The census statistics report of Country A warns that the country would suffer from brain drain, i.e. many residents with a high education level migrate to other countries.

- Explain **TWO** effects of brain drain on the economic growth of Country A. (4 marks)
- In the light of globalisation, suggest **ONE** policy the government of Country A could adopt to relieve the problem. Briefly explain your answer. (2 marks)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

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Question No. _____

Question No. _____

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Paper 1

Question No.	Key	Question No.	Key
1.	B (74%)	26.	D (65%)
2.	C (66%)	27.	C (57%)
3.	D (76%)	28.	B (64%)
4.	D (45%)	29.	A (41%)
5.	C (88%)	30.	C (74%)
6.	B (84%)	31.	D (65%)
7.	B (52%)	32.	A (76%)
8.	B (86%)	33.	*
9.	D (49%)	34.	D (34%)
10.	A (94%)	35.	A (57%)
11.	B (72%)	36.	C (61%)
12.	B (76%)	37.	B (65%)
13.	D (64%)	38.	*
14.	A (55%)	39.	A (69%)
15.	A (37%)	40.	C (34%)
16.	B (71%)	41.	B (60%)
17.	A (67%)	42.	C (56%)
18.	D (44%)	43.	B (48%)
19.	A (43%)	44.	A (52%)
20.	B (26%)	45.	D (56%)
21.	C (68%)		
22.	D (61%)		
23.	C (54%)		
24.	D (37%)		
25.	C (48%)		

* This item was deleted.

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

General note on item deletion

It is normal for the HKEAA to delete a small number of items from its multiple-choice question papers if they prove unsatisfactory. In practice, there are a number of reasons why this is considered necessary. By far the most common reason for deleting an item is that the item fails to discriminate between weak and able candidates – in other words, the majority of the candidates involved had to rely on guesswork in answering that question. If such an item is retained, the measurement process is rendered less effective. Where items have been deleted in the live papers, they are still included in this series of publications. They are indicated as deleted items. Such items may be discussed in the corresponding examination reports.

These documents were prepared for markers' reference. They should not be regarded as sets of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret their contents with care.

The answers provided in the marking scheme are for reference only. They are not the only possible answers. Alternative answers are acceptable so long as they are well reasoned.

The examination emphasises the testing of the understanding of economic theories and the application of the knowledge of economic analysis to practical problems. Candidates are advised to study this document in conjunction with the examiner's comments on candidates' performance in this booklet.

For essay-type questions, candidates are expected to demonstrate an understanding of the question, an ability to deploy relevant knowledge of the subject in response to the questions, and to present their answers logically and coherently.

In questions asking for a specified number of reasons or examples etc. and a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, and if a candidate gives three answers, only the first two should be marked.

The following symbols are used:

/	A single slash indicates an acceptable alternative within an answer.
@	The number in front of the symbol indicates the marks for each point.
max	Maximum mark for the question/sub-question

1. The full cost of buying the new model includes the price of new model (monetary cost) plus the value of time spent logging in the website. The resale value of the new model is not a forgone option so the cost of buying the new model does not change when such value falls.
- Marks
(3)

or
The drop in resale value will lower the (speculators') demand—hence, excess demand—for the new model, reducing the time spent logging in the website and thus the time cost of buying the new model. The cost of buying the new model falls.

or
(3)

2. (a) Public limited company/listed company.
- Marks
(1)

(b) Advantages:
– no interest burden
– no redemption obligation
[Mark the FIRST ONE point only.]

} @2
max: 2

Disadvantages:
– diluting the control right over the firm
– higher risk of being taken over
[Mark the FIRST ONE point only.]

} @2
max: 2

(c) Reasons:
– cost of advertising shared by a larger number of students
– bulk purchase of raw material in production
– more extensive division of labour
– Any other relevant point
[Mark the FIRST THREE points only.]

} @1
max: 3

3. The law states that (holding technology constant) when more units of a variable factor are added successively to a given quantity of fixed factors, the marginal products of this factor will eventually diminish.
Yes, when labour increases from 3 to 4 units, the marginal product of labour drops from 18 to 14 units, so the law of diminishing returns holds.

Marks
(3)

4. Taking photos may disturb other people in the restaurants/sharing the photos in the social media may disclose the design of dishes. Both possibilities may have an undesirable effect on the revenue/profit of the restaurants, but such effects are not compensated (financially or otherwise) by the photo-takers. Such photo-taking-and sharing behaviour may thus involve negative externality (external cost is involved).

Marks
(3)

or
Taking photos and sharing them in social media may have a promotion effect, attracting more customers—bringing higher revenue/profit—to the restaurant, but the photo takers do not receive compensation for such benefits. Such photo-taking-and sharing behaviour may thus involve positive externality (external benefit is involved).

or
(3)

5. When the price level increases, the adjustment of input prices is imperfect/incomplete (e.g. due to the long term contracts of factor of production), so the real cost of production —i.e. nominal cost divided by the general price level—will fall. Firms would thus use more factor inputs to produce larger output. Therefore, the SRAS curve is upward-sloping, i.e. a higher price level would result in a higher output level.

Marks
(3)

6. (a) The unemployment rate = $200\,000 / (3\,800\,000 + 200\,000) \times 100\% = 5\%$

Marks
(2)

- (b) Loss caused by unemployment:
– loss of output
– loss of human capital
– political and social unrest
– any other relevant point
[Mark the FIRST THREE points only.]

} @1
max: 3

- (c) Economic phenomena:
– rising real GDP growth rate
– rising inflation/lower deflation
– rising levels of investment/consumption
– any other relevant point
[Mark the FIRST TWO points only.]

} @1
max: 2

7. (a) In Country A, the opportunity cost of producing 1C = $6/2 = 3T$
The terms of trade: 1C will exchange for $(1/2 =) 5.5T$
As the opportunity cost of producing 1C is lower than the terms of trade, Country A will produce and export C.

Marks
(3)

- (b) The gain from trade = $5.5 - 3 - 2 = 0.5T$ per unit of C.

(3)

8. (a) Monetary base = \$500 million + \$1000 million = \$1500 million
Money supply = \$500 million + \$4000 million = \$4500 million

Marks

(1)

(1)

- (b) Change in monetary base = \$1500 million - \$1500 million = 0

(2)

Change in money supply

= New money supply - Original money supply

= (\$500 million + \$1000 million × 2) - \$4500 million = -\$2000 million

(2)

or

Change in money supply

= Change in cash held by non-bank public + Change in deposits

= 0 + (\$1000 million × 2 - \$4000 million) = -\$2000 million

or

(2)

- (c) (i) Under a fractional reserve banking system, banks are required to keep only a fraction of their deposits as reserves. They can create credit by lending out the remaining fraction in the form of loans to businesses and private individuals, which would eventually find their way back into the banking system in the form of deposits, resulting in a multiple increase in deposits/a banking multiplier larger than unity.

(3)

- (ii) If the legal reserve ratio is 100% (RRR=1) or if all of the banks decide not to lend out their excess reserves even when $RRR < 1$ (e.g. during financial crisis, when default rates are high), the money supply will be equal to the monetary base.

(1)

9. Illustrate in the diagram:

Marks

- Embargo effect: leftward shift of demand curve

(1)

- An original kinked supply curve S_0 , with the vertical segment at the quota level Q_0

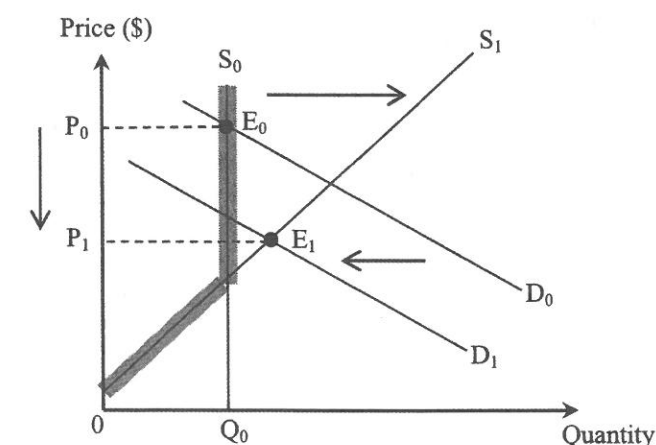
(1)

- Quota-abolition effect: A new upward-sloping supply curve, overlapping the kinked one below the original quota level Q_0

(1)

- Intersection of new demand curve (D_1) and new supply curve (S_1) results in a reduction in P

(1)



Section B

10. (a) Monopolistic competition.

Features:

- Many sellers
 - Price searchers
 - (slightly) differentiated products
 - Non-price competition
 - Imperfect market information
 - Any other relevant point
- [Mark the FIRST TWO points only.]

Marks
(1)

@1
max: 2

(b) Land. It is because it is a natural resource which is a gift of nature.

(2)

(c) Verbal elaboration:

- Condition: demand for solar panels is more elastic than its supply.

(2)

Indicate in the diagram:

- upward shift of the supply curve
- P increases
- Q decreases
- Correct position of buyers' burden and producers' burden
- buyers' burden < producers' burden

(1)

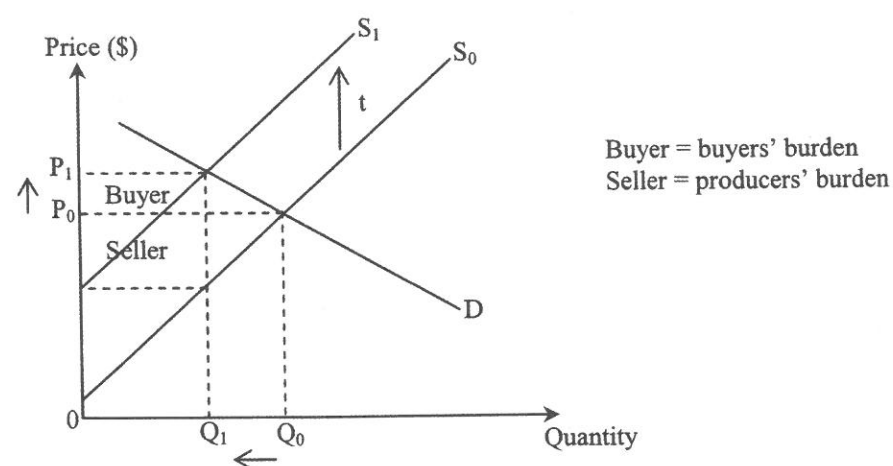
(1)

(1)

(1)

(1)

Chinese Solar panels sold in European Union



11. (a) No.

Marks
(1)

because public services are usually subsidized—i.e. provided at or below cost, which is in turn lower than market price. Their contribution to GDP (or market value), computed using market price, would thus be higher than \$45 (the subsidized price charged by the clinic).

(2)

or

because the public services which are not sold at the market price will be counted using factor cost in the GDP, which is higher than \$45.

or

(2)

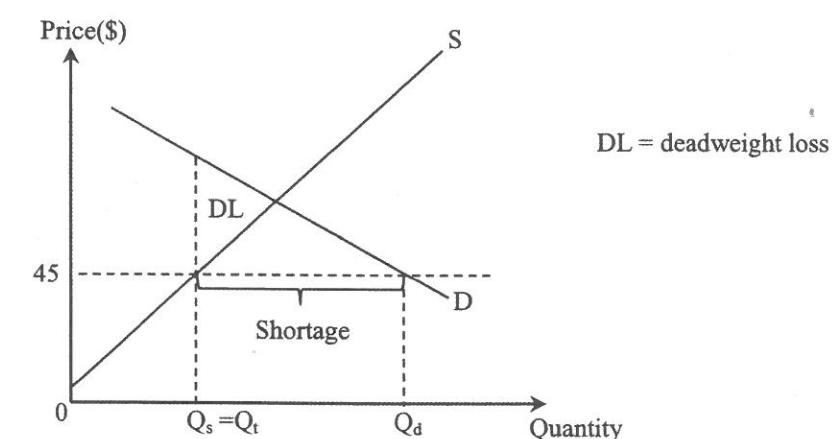
(b) Illustrate in the diagram:

- Price set below equilibrium¹
- Correct position of Q_t
- Correct position of deadweight loss

(1)

(1)

(1)



Verbal elaboration:

- Price is set at \$45 which is below equilibrium.
- Quantity transacted is below equilibrium quantity or there is underproduction or shortage.
- Marginal benefit > Marginal cost

(1)

(1)

(1)

(c) Equalize income/outcome: basic medical service acts like a transfer or subsidy-in-kind, enjoyed mainly by the low-income group; The high-income group may choose not to "queue up" for such medical benefits and turn to private medical service instead.

(2)

or

Equalize opportunities: basic medical service provided to patients with chronic diseases would improve their health, allowing them to continue to work and earn income. Therefore, their income would not be affected by their ability to afford medical service (their background), but rather by their work effort.

or

(2)

¹ The difference between the terms "equilibrium" and "market clearing" is not required.

- (d) Equity:
After the fee raise, the poor may no longer be able to afford the fee and thus be deprived of the opportunity to enjoy the medical services. Equity may be worsened as the poor cannot enjoy the transfer/subsidy while the rich can still use the service. (2)

Or

After the fee raise, the poor may no longer be able to afford the fee and thus be deprived of the opportunity to enjoy the medical services. The poor may then be deprived of the chance to work and earn income due to their poor health. (2)

Efficiency:

The fee increase would improve efficiency as it results in a price closer to the equilibrium level, thus narrowing the gap between MB and MC and reducing the size of deadweight loss. (2)

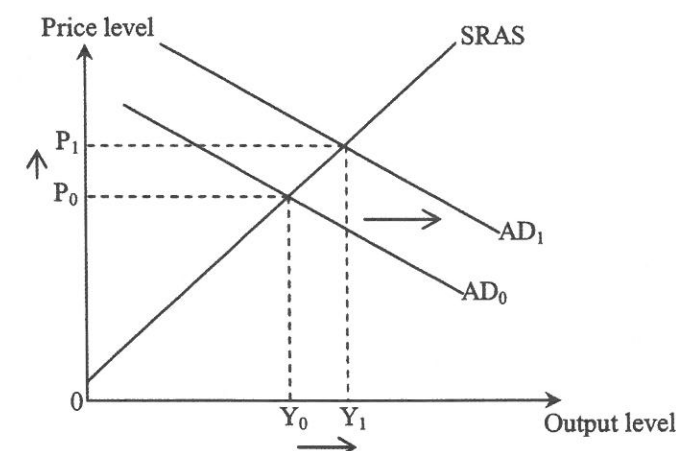
12. (a) The relative importance of the gaming industry has been declining because its gross revenue has been falling faster than GDP at current market price. (2)

Marks
(2)

(b) Verbal elaboration:

- The cash-sharing scheme would raise the disposable income of the residents (1)
- and therefore result in an increase in consumption. (1)
- AD would rise, (1)
- raising price and output levels accordingly. (1)

Illustrate in the diagram:



- the rightward shift of AD (1)
- higher P and Y (1)

(c) Arguments for:

- Investment would increase potential output over time, but cash sharing would not have similar long-run effects on the economy.
- Investment would raise AD directly. But the cash-sharing scheme could only raise AD indirectly via an increase in consumption. In case the residents choose not to spend the cash received on consumption, the AD effect would vanish.
- Any other relevant point

@2
max: 3

Arguments against:

- During economic downturn, a reduction in social welfare due to suspension of the cash-sharing scheme may result in discontent of the citizens and political instability.
- While people can benefit from the cash-sharing scheme instantly, it may take a longer time for investment to boost their income.
- Public investment may be inefficient due to high administrative costs or low sensitivity to market/price signals (lack of profit incentives).
- Any other relevant point

@2
max: 3

Remark: Maximum mark of part (c) is 5 marks

13. (a) Division of labour/specialization

Reasons:

- practice makes perfect
 - choosing the most suitable person for the job
 - save time in training
 - stimulus to mechanisation
 - Any other relevant point
- [Mark the FIRST THREE points only.]

Marks

(1)

@1
max: 3

(b) (i) Proposal A on current account:

an increase in income tax would lower disposable income and result in a drop in consumption, AD, as well as output level. Import would fall, implying a bigger surplus or smaller deficit in the current account balance.

(1)

(1)

(1)

(1)

max: 6

Proposal B on current account:

A fee charged on departing travellers would result in an increase in the cost of using the airport.

Travelling industry would be adversely affected, reducing the number of both incoming and outgoing travellers.

Both import and export of services would drop, causing an ambiguous change in the current account balance.

(1)

(1)

(1)

(1)

(ii) Proposal A on average flight distance:

After the tax increase, some poor people would be unable to afford long-distance flights and would switch to short-distance flights, thus reducing average flight distance.

or

The increase in income tax would not affect the relative price between short-distance flights and long-distance flights, so consumption of the flights would not be affected.

or

After the tax increase, some poor people would be unable to afford the flights. As budget-constrained people would often take short-distance flights, the drop-out of these travellers would result in an increase in average flight distance.

@2

max: 2

Proposal B on average flight distance:

A lump-sum fee charged on both long-distance and short distance travellers will result in a drop of the price of long-distance flight relative to short-distance flight. A higher proportion of travellers will thus take long-distance flights, increasing the average flight distance.

(4)

Remarks: Maximum for (b) is 10 marks.

Section C

14. (a) (i) - Correct position of Q_m
- Correct position of P_m

Marks

(1)

(1)

- (ii) Illustrated in the diagram:
correct position of DL_0

(1)

Verbal elaboration:

At Q_m , marginal benefit (MB) is higher than marginal cost (MC).

(1)

- (b) (i) The reduction in fuel cost reduces the marginal cost.

(1)

- (ii) Verbal elaboration:

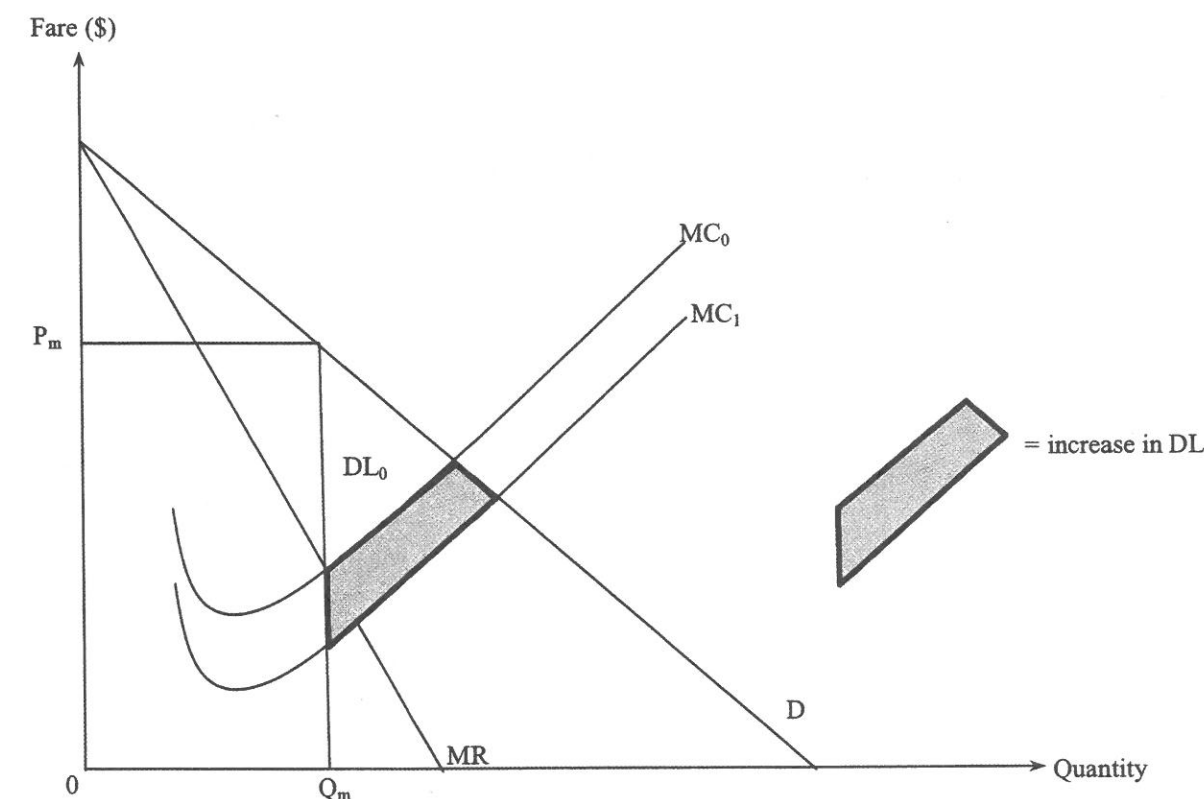
Economic efficiency would be lowered and the deadweight loss would be larger, as the gap between MB and MC would be enlarged and output would now be even further away from its efficient level.

(2)

Illustrated in the diagram:

correct position of new DL/increase in DL/the increased gap between MB and MC

(1)



(c) Yes. As two groups of consumers (adults and children) are charged different prices for the same good (ferry service) produced at the same cost. (3)

(d) (i) If owners of the seafood restaurants agree to engage in price fixing and raise the price together (i.e. collusion), they would violate the first conduct rule under the Competition Ordinance. (3)

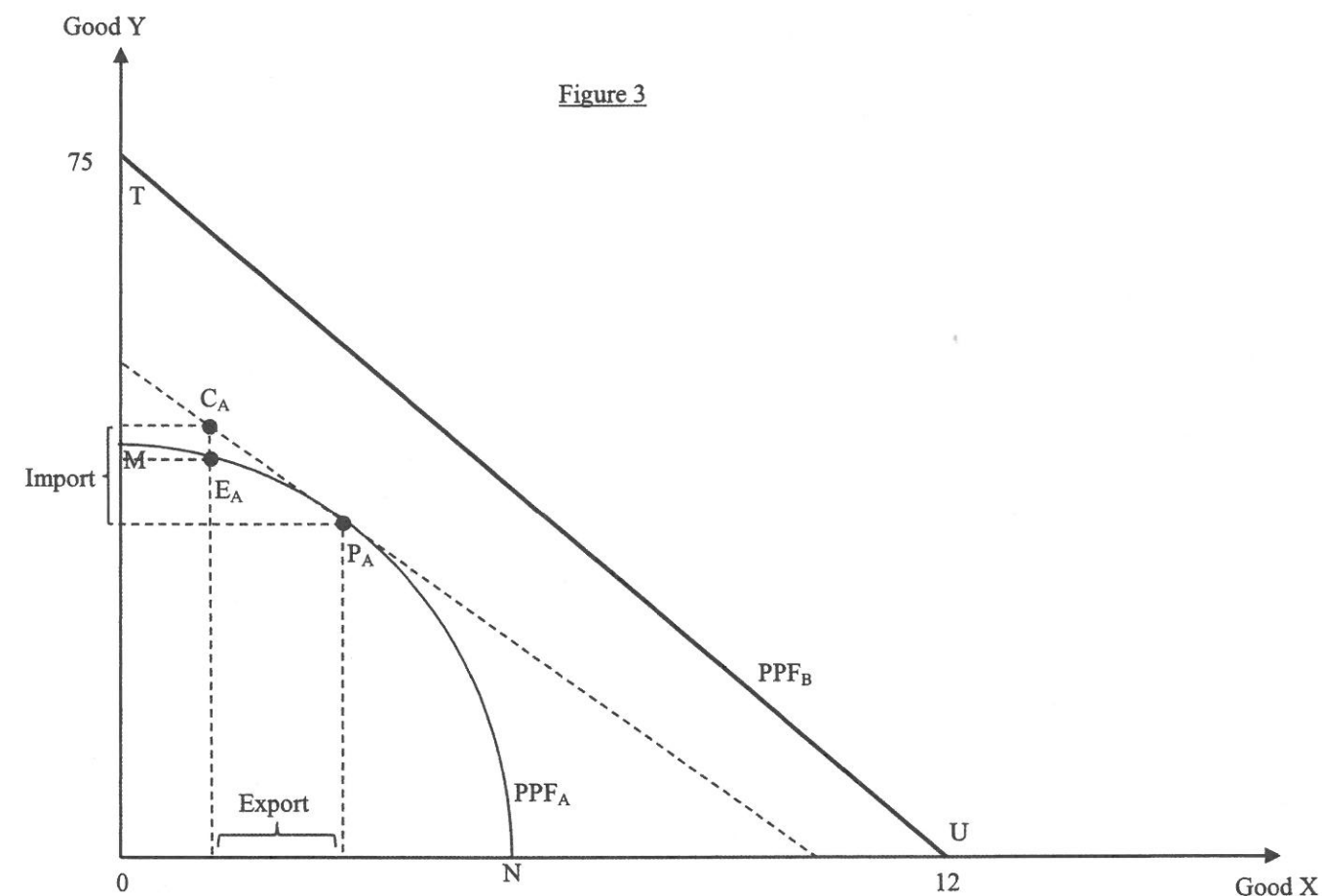
(ii) The price increase may be a market response to increased demand for seafood meals during public holidays due to a bigger number of visitors. (2)

or

The price increase may be a result of reduced supply of seafood meals due to higher production costs (e.g. need to hire more part-time workers) on public holidays. (2)

15. (a) The marginal costs of producing of both X and Y are increasing in Country A. (1)

(b) Country A's opportunity cost of producing $1X = 4Y$ (1)
Country B's opportunity cost of producing $1X = 6.25Y$ (1)
Country A has a lower cost in producing X, and thus it has a comparative advantage in producing X. (1)



Candidates' Performance

Paper 1

Candidates' overall performance was good. The mean score of this paper is 26 out of 43. The following questions deserve special attention.

Q.15 A decrease in the cost of producing Good X results in a 5% and 10% change in its price and quantity transacted respectively. Which of the following best explains the above changes?

- *A. Good X is a durable good. (37%)
- B. There are no close substitutes for Good X. (20%)
- C. Firms producing Good X have excess capacity in production. (32%)
- D. The factors of production for Good X are not easily available. (11%)

This question tests candidates' understanding of factors affecting demand and supply and their elasticities. Many of them seemed to understand that either/both of the demand for and supply of Good X is/are elastic because the percentage change in the quantity transacted is bigger than that in the price. However, some candidates failed to see that the fall in production cost had resulted in a rightward shift of the supply curve, so the corresponding changes in price and quantity transacted represented a movement along the demand curve—reflecting the responsiveness of consumers rather than producers.

Q.20 Which of the following statements about monopoly are correct?

- (1) A monopolist sets the market price.
- (2) A natural monopoly arises when no firms are permitted to enter the market.
- (3) There are substitutes for the products provided by a monopolist.

- A. (1) and (2) only (42%)
- *B. (1) and (3) only (26%)
- C. (2) and (3) only (11%)
- D. (1), (2) and (3) (21%)

This question is about monopoly. Some candidates did not seem to understand that the monopoly power behind a "natural monopoly" lies in the firm's cost advantage, rather than government regulations. Other candidates did not realise that although the monopolist is the sole producer of its own good, it may still face potential competition from other firms producing similar goods.

- (c) Indicate in the diagram:
 - (i) The correct position of P_A (1)
 - (ii) The correct position of C_A (1)
 - (iii) The correct volume of export (1)
 - The correct volume of import (1)

- (d) No, because the opportunity cost of producing X (and Y) will be unaffected by this technological change. (2)

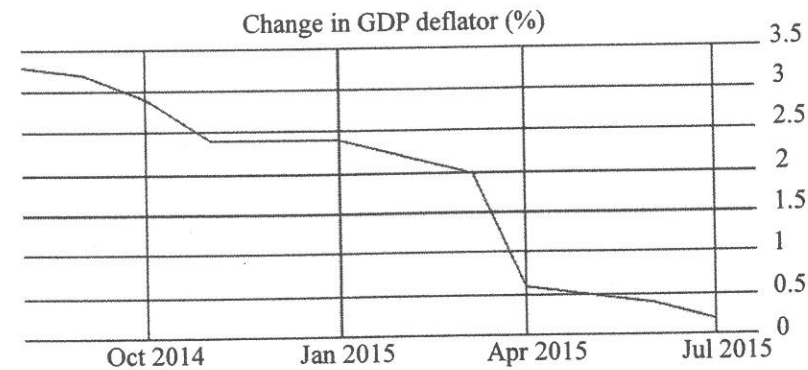
- (e) (i) Effects on economic growth:
 - Lower per-capita GDP
 - Lower human capital/labour productivity
 - Lower marginal productivity of capital (due to complementarity of labour and capital)
 - Slower research and development
 - Any other relevant point
 [Mark the FIRST TWO points only.]

} @2
max: 4

- (ii) Policy:
 - Importation of high-skilled labour
 - Scholarships to attract higher-quality overseas students
 - Relaxation of immigration policy to target the smarter brains
 - Introduction of return migration policy
 - Any other relevant point
 [Mark the FIRST point only.]

} @2
max: 2

Q.24 Study the diagram below.



The diagram shows that during the period from October 2014 to July 2015,

- A. the nominal GDP was falling. (15%)
- B. the purchasing power of money was increasing. (23%)
- C. the growth rate of real GDP was higher than that of nominal GDP. (25%)
- *D. the nominal interest rate was higher than the realised real interest rate. (37%)

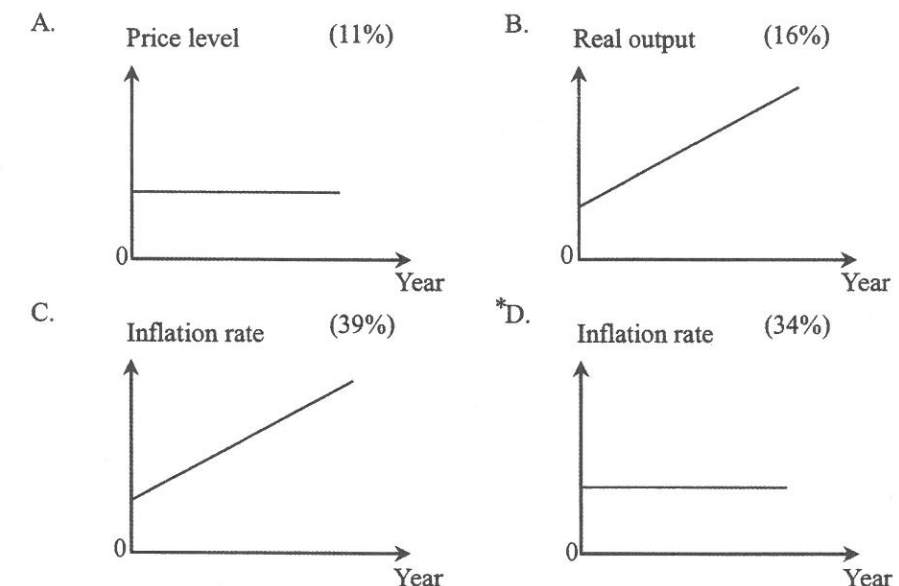
This question tests candidates' ability to extract relevant information from the data/figure to identify the correct economic phenomenon. Nearly half of the candidates confused disinflation (the case here) with deflation, hence picking B or C as their answer. To get the correct answer, candidates had to understand that the difference between the nominal interest rate and the actual (realised) inflation rate is the realised real interest rate.

Q.33 In a fractional reserve banking system, the actual deposit created is often below the maximum possible deposit created. The gap between these two amounts will be narrowed when

- A. the popularity of electronic payment increases.
- B. the central bank tightens the restrictions imposed on mortgage loans from banks.
- C. the central bank buys bonds from the public.
- D. the central bank reduces the legal reserve ratio for banks.

This item was deleted as it did not effectively assess candidates' knowledge and understanding. This question tests simultaneously candidates' knowledge of the creation of both (i) actual deposits and (ii) maximum-possible deposits as well as the change in their relation in different cases. Maximum-possible deposit is calculated under the assumptions of no cash leakage and no excess bank reserves—the "benchmark". Many candidates seemed to understand that while an expansion in the monetary base (option C) or a reduction in the legal reserve ratio (option D) would increase both (i) and (ii), the tightening of loan restrictions (option B) would only reduce (i) without altering (ii). Following the same logic, candidates might have concluded that people holding less cash for transaction purpose (option A) would only raise (i) without affecting (ii). Such conclusion, however, overlooked the implicit assumption of zero cash leakage in the benchmark and the fact that reserves in the bank may be altered by the cash holding of the public.

Q.34 According to the classical quantity theory of money, which of the following diagrams is correct when money supply is growing at a constant percentage each year?



This question is a test of the classical quantity theory of money (QTM) via graphical representation of economic data. Many candidates were able to apply the QTM in the given case to deduce that a growing money supply would result in a rising price level, i.e., a positive rate of inflation. However, quite a number of candidates did not realise that, under QTM, the price level should be increasing at a constant rate, meaning that the inflation rate should be constant over time. Some might have mis-read the label of Y-axis ("Inflation rate") in option C as "Price level" and thus mis-conceived the upward-sloping straight line as representing a constant inflation rate. Candidates should be more cautious about the distinction between price level and inflation rate (i.e., percentage change in price level over time).

Q.38 If the government simultaneously increases its expenditure and income tax by the same amount, the effect on the aggregate output is _____.

- A. contractionary
- B. neutral
- C. expansionary
- D. indeterminate

This item was deleted as it did not effectively discriminate between weak and able candidates. This question examines the combined effect of an equal increase in government expenditure and income tax on aggregate output. Candidates might have overlooked the possible effect of the tax change on aggregate supply. In particular, the increase in the income tax may have a disincentive effect on labour supply, which would then lower aggregate supply. Combining the effects on aggregate demand and aggregate supply would result in an indeterminate change in aggregate output.

Paper 2

Section A

Q. number	Performance in General
1	Fair. Most candidates managed to define opportunity cost correctly, but some of them failed to relate it to the given case. Many misperceived the change in resale value of the new model as a change in monetary cost, or they confused the chosen option with the forgone option in this case. Some candidates did not notice that both monetary cost and non-monetary cost were involved in buying the new model, and thus failed to apply the concept of full cost to this case.
2(a)	Excellent.
2(b)	Good. Ignoring the key words “from the viewpoint of owners,” some candidates gave totally irrelevant answers. A minority of candidates did not indicate clearly whether a particular argument is an advantage or a disadvantage, resulting in scoring fewer marks.
2(c)	Satisfactory. A majority of candidates were able to provide good reasons for economies of scale. A minority of candidates only spelt out some terms (e.g. marketing economies) or phrases (e.g. resources are fully utilised) without elaborating with reference to the given situation in the question or without relating to the reduction in average cost.
3	Excellent. A small number of candidates wrongly applied the concepts of “fixed cost / variable cost / average product”.
4	Satisfactory. The following are common mistakes: – failure to state whether the externality was positive or negative; – giving confusing arguments without specifying who got the benefit and who needed to be compensated.
5	Satisfactory. Many candidates were aware that the incomplete/slow adjustment of input prices was the key reason. However, some failed to relate their argument to the price level and output level in their elaborations: they neither started their argument from a change in general price level, nor did they reach the conclusion of a change in output level at the end. A few candidates wrongly used the terms price (rather than the price level) and quantity supplied (rather than aggregate output) in their explanation, showing either their confusion of supply-demand model with AS-AD model or their carelessness.
6(a)	Good. A few candidates wrongly included the underemployed population in their calculation.
6(b)	Satisfactory. A common mistake was to regard the rise of unemployment benefits—a transfer payment—as a loss to society.
6(c)	Excellent. A small number of candidates did not read the question carefully, and included “the fall in unemployment rate” as part of their answers.
7(a)	Satisfactory. Many candidates compared correctly the costs of domestic production of Country A with the terms of trade to determine the type of good exported by Country A. Common errors for those who failed to get the answer: – miscalculation of the opportunity cost from the input-requirement data; – mistakenly comparing the unit costs of producing the two goods directly in terms of their input requirements.
7(b)	Satisfactory. Most candidates were able to show their knowledge about the gain from trade in the case of positive transportation cost.

Q. number	Performance in General
8(a)	Excellent.
8(b)	Satisfactory. Most candidates used correctly the new required reserve ratio (RRR) in their calculation. However, some of them simply calculated the new levels of money supply and monetary base without finding their changes (relative to their old levels). A few of them were not given full credit because they did not show their workings or omitted the units. Weaker candidates failed to work out the case of change in RRR with constant reserves.
8(c)(i)	Fair. Many candidates were able to show their basic understanding of credit creation, but only some of them stated precisely the underlying assumptions. Others tried to apply the formulae for monetary base and money supply (which deserves some credit), but very few candidates were able to relate the difference between the two to the fractional reserve system (i.e. $RRR < 1$).
8(c)(ii)	Satisfactory. Many candidates were able to state that credit creation would not work when $RRR = 1$, or when banks do not make loans and hold all deposits as reserves.
9	Good. A majority of candidates correctly showed the effect of an embargo by shifting the demand curve leftward. Some also demonstrated a good knowledge of quota by drawing the kinked supply curve. However, a few of them misunderstood the key word “abolish” as meaning introduction of a quota, and thus reached a wrong conclusion. Some candidates also failed to represent the equilibrium price and quantity at the intersection of demand and supply curves. A minority of candidates mistakenly used the small open economy model to address this large country question.

Section B

Q. number	Performance in General
10(a)	Good. Many candidates stated correctly the market structure of the industry. However, some candidates wrongly focused on the top 10 manufacturers (whose market shares were actually rather small) and gave oligopoly as their answer. Confusing terms such as “monopoly competition”, “monopolistic” and “競爭性壟斷” were found in some of the answers.
10(b)	Good. A minority of candidates wrongly regarded sunlight as capital. Some mis-read the question and went on to explain the type of factor inputs for producing PV modules or to explain the type of production of the solar energy industry.
10(c)	Good. Most candidates were able to draw the diagram with labels showing the correct positions of the tariff burdens. However, unclear concepts were revealed in some candidates’ verbal explanation. Some only stated elastic demand as the condition, but failed to relate the tariff burden to the difference in the elasticities of demand and supply. A few candidates mis-used the model of small open economy (though partial credits were still given). A minority wrongly discussed the distribution between consumer and producer surpluses.
11(a)	Poor. Many candidates misinterpreted the question as meaning “whether the \$45 should be included in the calculation of GDP”—a frequently asked question in past papers. Some over-emphasised the imported raw materials (medicine) and reached the wrong conclusion of “less than \$45”. Among those who got their stand correct, only some of them provided a good explanation.
11(b)	Satisfactory. Most candidates managed to draw a well-labelled diagram for the case of excess demand, while many were able to indicate the position of deadweight loss. However, only a few of them were able to accurately state that the inefficiency was caused by $MB > MC$ in this case. A minority of candidates represented their answer in a diagram of marginal social cost and marginal private cost, which was neither required under the existing curriculum and assessment framework nor applicable to this question.

Q. number	Performance in General
11(c)	Good. Most candidates were able to give an explanation in their own words, but only some made use of the concepts of “equalising income/opportunity” in their answers.
11(d)	Good. Many candidates were able to demonstrate their understanding of the effects of the fee increase on efficiency and equity, and some could explain the efficiency-equity trade off in this case as well. The following shows a list of mistakes: – thinking that they were only required to provide reasons to support the argument; – mis-labelling their statement about efficiency under the heading of equity, or vice versa; – failure to provide an explanation for their stand using economic terms.
12(a)	Poor. Many candidates were able to identify the trends of the income of the gaming industry and the GDP of Macau, but failed to relate those data to the relative importance of the gaming industry. Among those who got the trend correct, only some gave accurate explanations based on the data. Instead, some others suggested reasons for the drop in the revenue, such as a reduction in tourist numbers.
12(b)	Excellent. Most candidates provided a systematic explanation for the effect of the cash-sharing scheme on price and output levels with the aid of a well-labelled diagram. A minority of candidates mistook a rise in transfer payment for a rise in government consumption expenditure, or an increase in money supply. Others confused disposable income with taxable income in their answers.
12(c)	Poor. Most candidates were able to either support or oppose the economist’s suggestion in economic terms. However, only some managed to provide a balanced analysis by including both the pros and the cons. The following problems were identified: – presenting argument(s) exclusively for or exclusively against the suggestion; – failure to compare the extent of change in AD due to the change in investment with that due to an equal-amount change in transfer payment; – mistakenly relating investment to a higher profit tax and cash-transfer to a higher sales tax, and then putting forward arguments in terms of the effects on the budgetary position of the Macau government.
13(a)	Excellent. Only a minority of candidates mistakenly regarded efficient use of capital goods and saving training cost as reasons for higher labour productivity.

Q. number	Performance in General
13(b)	Fair. Most candidates attempted to identify the effects of the two proposals, but only some demonstrated their ability to apply economic concepts and theories correctly in this case. The following shows a list of mistakes for both parts that deserve attention: – comparing the effects of building and not building the additional runway rather than the effects of the two proposals to finance the construction of the runway; – mis-reading the question as meaning “how proposal A affects the current account” and “how proposal B affects the average flight distance”; – poor organization of answers, without a clear indication as to which of their arguments refers to which proposal. Part (i): For proposal A, the performance was fair. Most candidates simply stated that the increase in income tax would lower disposable income and consumption, while the stronger candidates were able to relate the fall in consumption to a fall in output as well as imports, and ultimately the current account. For proposal B, many candidates did not realise that the expenditure/revenue of the tourist industry was counted as import/export of services and that the rise in the cost of using the airport would adversely affect both foreign visitors and local people flying to other countries. Some candidates gave confusing answers without indicating whether they were talking about incoming or outgoing travellers. Part (ii): For proposal A, the performance was fair. Some candidates showed their ability to apply concepts and theories to an unfamiliar situation. Others only stated their stand (increase / decrease / no change) without proper reasoning. For proposal B, the performance was satisfactory. Most candidates could apply the law of demand to deduce that the average flight distance would increase. Although many were able to use the concept of relative price, only some of them were able to give a complete and logically consistent explanation. Common mistakes include: – inability to state, in words or in formulae, that the same lump-sum fee was charged on the two types of flights with different flight distances and airfares; – stating that the relative price of long-distance flights would become lower than that of short-distance flights; – failure to understand that the fall in the relative price of long-distance flights would lead to a rise in the proportion of such flights (relative to short-distance ones) taken by departing travellers without necessarily increasing their total quantity.

Section C

Q. number	Popularity	Performance in General
14(a)(i)(ii)	53%	Good. Most candidates were able to indicate correctly the output and price and explain why deadweight loss exists under monopoly pricing. A minority of candidates did not use the symbols Q_m and P_m as required.
14(b)(i)(ii)		Candidates' performance in part (i) was excellent. However, the performance in part (ii) was fair, as some misconceived an upward shift in the marginal cost (MC) curve as a reduction in marginal cost. Among those who were able to draw the correct MC curves, some failed to give a proper explanation for the increase in the deadweight loss.
14(c)		Good. Most candidates performed well in this question about price discrimination (PD). Weaker candidates only recited the definition of PD without relating it to the given case, or simply stated it as an example of third degree PD.
14(d)(i)(ii)		Satisfactory. In part (i), most candidates could identify the type of anti-competitive behaviour and correctly stated that it is a horizontal agreement or that it violates the first conduct rule. Some candidates did not provide reason(s) for the change in demand /supply in part (ii).
15(a)	47%	Satisfactory. However, some candidates gave incomplete answers, such as "resources are heterogeneous" or "products are heterogeneous".
15(b)		Excellent.
15(c)(i)(ii)(iii)		Satisfactory. Most candidates were able to indicate the correct position of P_A , but only some actually knew that Country A's import and export volumes were represented by the gap between its production point and consumption point for Good Y and Good X respectively.
15(d)		Good. Many candidates gave a good explanation for the unaffected trade direction. A minority of candidates did not indicate their stand.
15(e)(i)(ii)		Satisfactory. In part (i), most candidates could explain clearly how brain drain might slow down economic growth via different growth channels. In part (ii), some candidates misunderstood the key words "brain drain" and "in the light of globalisation", and thus gave irrelevant answers. Candidates were expected to explain how their policy proposal can offset the reduction in labour productivity or human capital through interactions with the rest of the world.

General comments and recommendations

1. Some candidates lack a thorough understanding of basic economic concepts and principles, especially when applying them to solve hypothetical or real-life problems. Candidates should strengthen their analytical skills instead of simply regurgitating concepts and theories.
2. Some candidates do not always read the questions carefully. Candidates should devote special attention to the assumptions and conditions laid down in the questions to avoid giving irrelevant answers.
3. Some candidates have difficulty in presenting their answers in a precise way. Candidates should equip themselves with better language abilities as well as graphical skills.
4. Some candidates lack awareness of current economic issues in our society. Candidates should pay more attention to economic news.
5. Some candidates lack skills to extract information from tables and graphs as well as to draw well-labelled diagrams. Candidates should pay special attention to the headings, labels and axes as they are essential to our understanding of the data presented therein.

Acknowledgements

Material from the following publications/web-sites has been used in question papers in this volume:

Renewable Energy Policy Network for the 21st Century (REN21)	Renewables 2013 Global Status Report http://www.ren21.net/Portals/0/documents/Resources/GSR/2013/GSR2013_lowres.pdf
The Gaming Inspection and Coordination Bureau Government of Macau SAR	Monthly Gross Revenue from Games of Fortune in 2015 and 2014 http://www.dicj.gov.mo/web/en/information/DadosEstat_mensal/2015/index.html
Statistics and Census Service Government of Macau SAR	Gross Domestic Product (GDP) and per-capita GDP http://www.dsec.gov.mo/PredefinedReport.aspx?ReportID=32
South China Morning Post	Macau government tightens its belt after 15th consecutive month of falling casino revenues http://www.scmp.com/news/china/economy/article/1854603/macau-government-tightens-its-belt-after-15th-consecutive-month

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